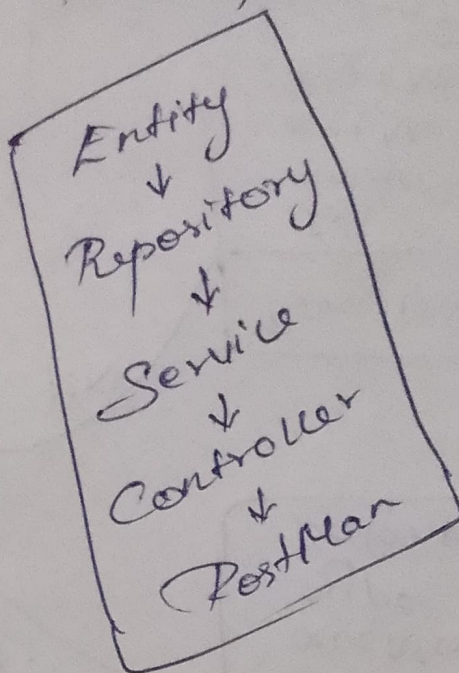


Week - 2

Banxico

- ① Verify project, dependencies, structure, entity creation & Hibernate
- ② Repo, DTOs, validation & Exception
- ③ Customer Module
- ④ Account Module
- ⑤ Transaction
- ⑥ Beneficiaries



Day - 1

Verifying Week 1

Separate Branch in git & Verifying dep,
postgres SQL

Entity Structure using Lombok

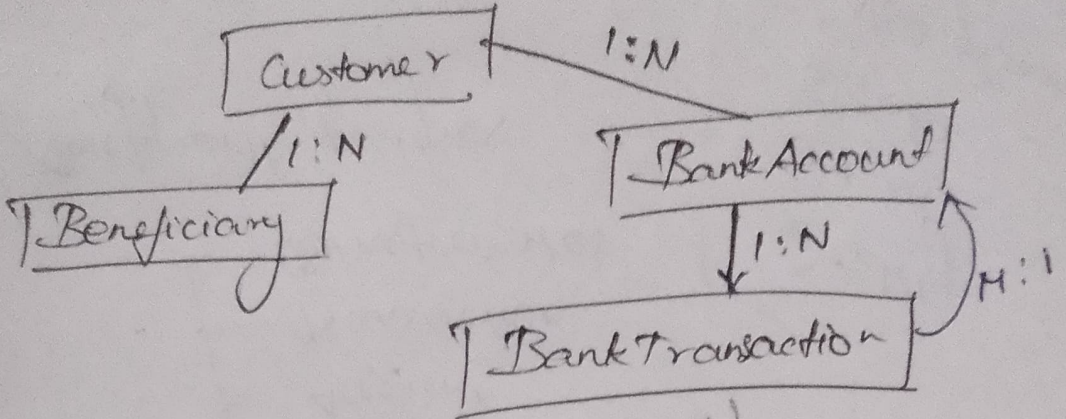
```
├── Customer
├── Account
├── Transaction
└── Beneficiaries
```

Verify with postgres
Schema

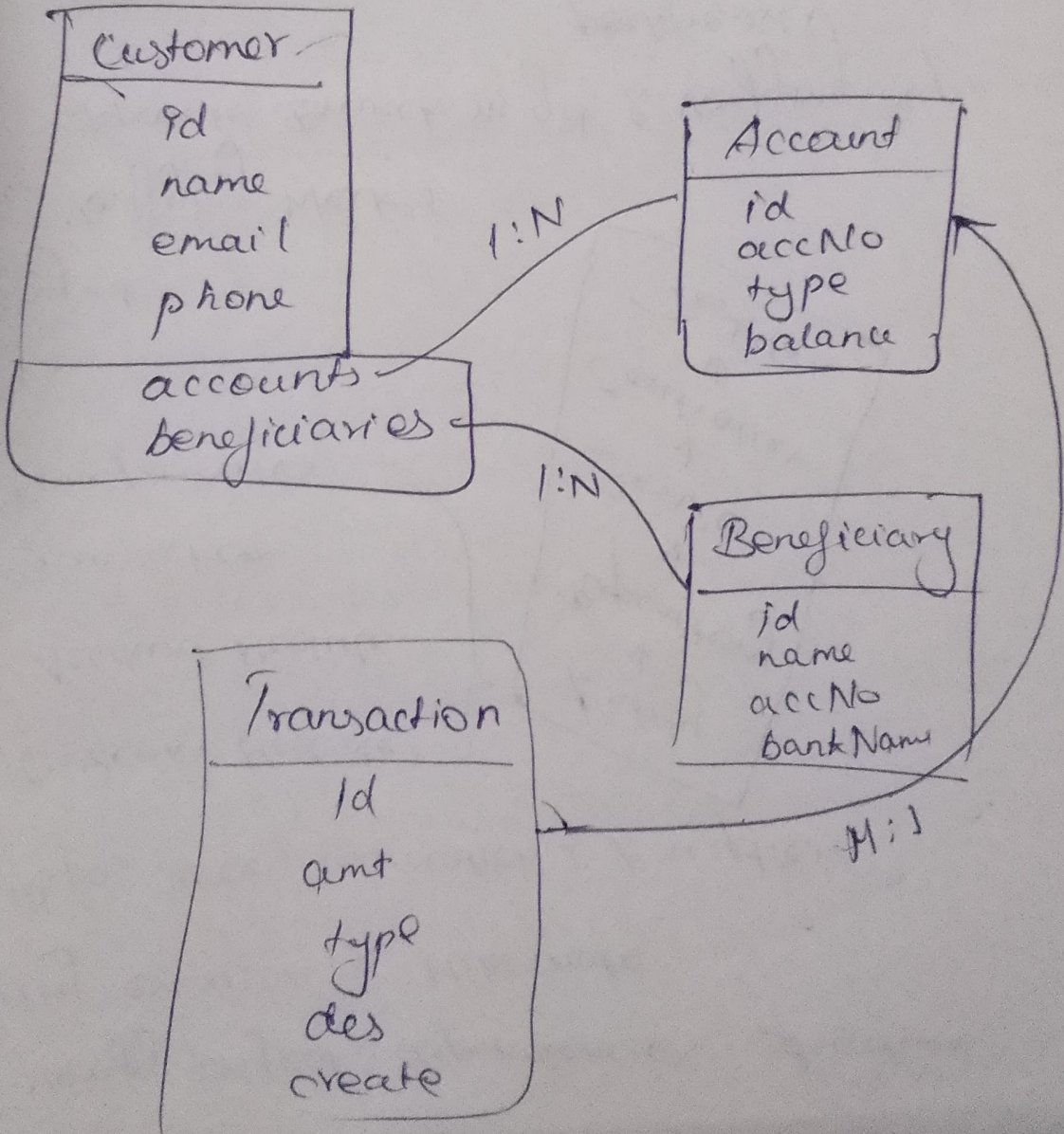
@Entity
@Table
@Data
@NoArgsConstructor
@AllArgsConstructor

Day 2

Relationships



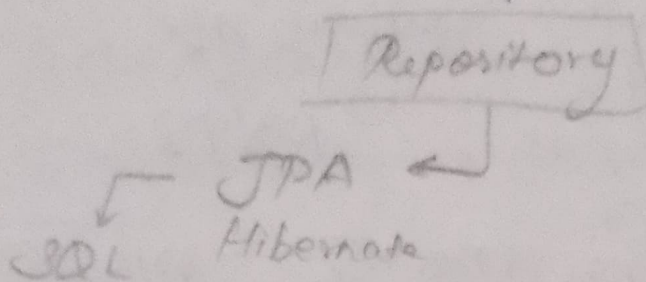
@OneToMany (mapped by = " ")



Automatically generates the SQL

Repository ^{Interface}

Controller $\rightarrow$ Service $\rightarrow$



Relationships
Repository
DTO
Validation
Exception

Importing

Entity
Repository
stereotype

save()
findAll()
findById()
deleteById
existsById
count

DTO

call $\rightarrow$ DTO $\rightarrow$ Entity $\rightarrow$ DB

<Entity, Primary Key>

Customer

name
email (*)
phone

Account

No
Type
Balance
cus_id

customer Id
 $\downarrow$
Entity

Transaction

Amt
Type
des
acc_Id

Beneficiary

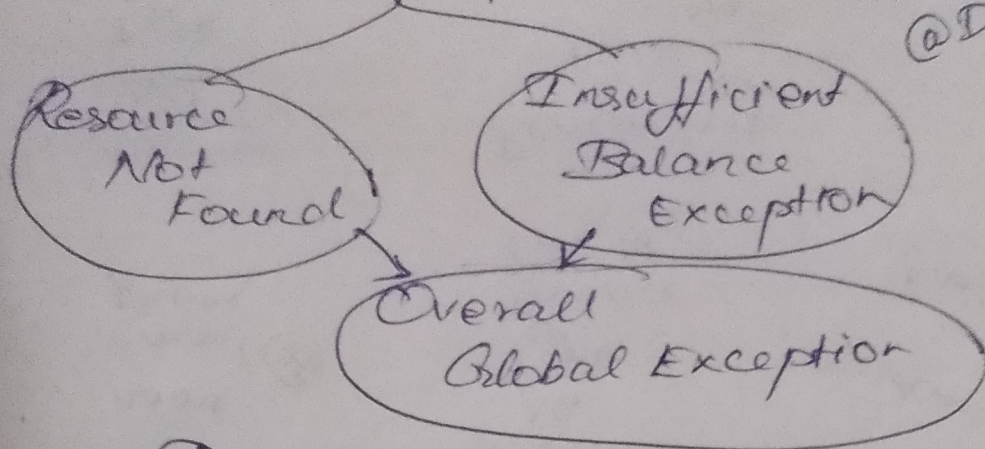
name
acc-No
B-name
cus_Id

Validation

DTO → Validation

Exceptions

Runtime Exceptions



@NotBlank → String
@Email
@NotNull → Obj
@DecimalMin

Returns

response entity

```
graph LR; A[response entity] --> B[Body]; A --> C[Status Code];
```

Day 3 Customer Module

Controller

↓

Service

↓

Repository

Service

Controller

Exceptions

Test

Get All

Get By Id

Storing Testing

@Autowired - obj automatically

Service - Business logic

Object

All Customer

Id

Postgres stores Entity

Controller - Receives the HTTP req

@RestController

@RequestMapping

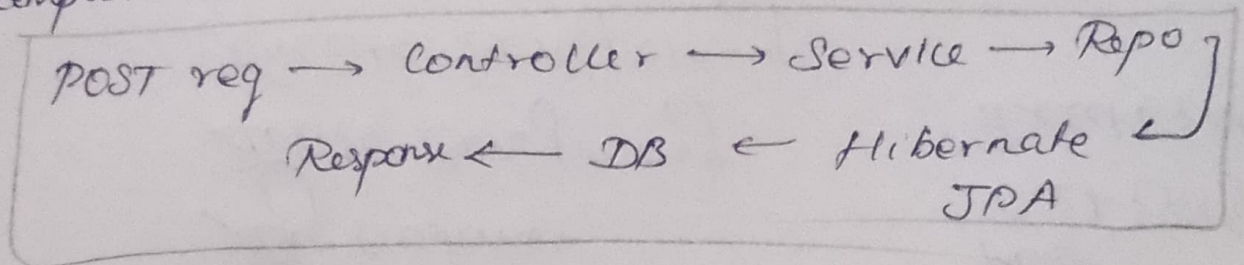
@PostMapping

@GetMapping

instead of

@RequestBody → JSON to Entity

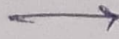
Sample Flow



Method Argument

Not Valid

Exception



Day 4 - Account

Service

Repo ← Bank Acc
Customer } contains many

Get All Account

Get By Id

Response Entity

Controller

/api/accounts

@PostMapping

@GetMapping

@GetMapping
("id")

Transaction

Credit

Debit

Balance

Service

→ Create Account

Credit Debit

→ find by id

→ History

Controller

Get
by Id

Post
By Id

@PostMapping ^{variable}
("/{accountId}/transactions")

@GetMapping
("/{accountId}/transaction")

Beneficiary

Service

All Beneficiaries

By customer

Delete Beneficiary

Create Beneficiary

Controller

Get

Post

Get {id}

Delete {id}